

Feature database documentation

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1 Overview

This document outlines the structure, naming conventions, and mathematical formulations used to construct the sliding-window feature database. The dataset is derived from electromyography (EMG) grids, inertial measurement units (IMUs), and goniometer signals.

Data is segmented into discrete windows of $N = 410$ samples (200 ms at 2048 Hz) with a 50% overlap (step size of 205 samples).

2 Database structure and naming conventions

The final database is a flat, 2D matrix (Pandas DataFrame) where each row represents a single 200 ms time window. The columns are strictly divided into three categories: Metadata, Features, and Targets.

2.1 Metadata

Used for cross-validation splitting and state tracking. These columns **must not** be used as training features.

- **Subject:** Unique identifier for the participant (e.g., '01', '09').
- **Timestamp:** The timestamp corresponding to the *end* of the sliding window.
- **Activity:** The discrete physical task being performed (used as the classification target).
- **Gait & Reps:** Additional contextual states at the end of the window.

2.2 Feature columns

Extracted statistical summaries of the sensor data within the window.

- **EMG features:** Named as `EMG_[Muscle]_[Metric]_Mean`. For example, `EMG_TA_WL_Mean` represents the spatial mean of the waveform length across the Tibialis Anterior grid.
- **IMU features:** Named as `IMU_[Axis]-[Location]_[Metric]`. For example, `IMU_aX-f_Jerk` represents the jerk of the X-axis accelerometer on the foot.

2.3 Target columns

Ground truth joint angles from the goniometers. Named as `G_[Joint]_Target`. For example, `G_DA_X_Target` represents the mean angle of the dominant ankle during the window.

3 Mathematical formulations

Let N be the number of samples in a window ($N = 410$). x_i represents the amplitude of a specific channel at sample i .

3.1 EMG Features

To prevent destructive interference (phase cancellation), features are calculated independently for each of the $M = 64$ channels in a muscle grid, and then spatially averaged to represent the muscle activation. $x_{c,i}$ is sample i of channel c .

1. Mean absolute value (MAV) Measures the overall amplitude and power of the muscle contraction.

$$\text{MAV} = \frac{1}{M} \sum_{c=1}^M \left(\frac{1}{N} \sum_{i=1}^N |x_{c,i}| \right) \quad (1)$$

2. Waveform length (WL) Captures both the amplitude and frequency of the signal by measuring its cumulative length over time.

$$\text{WL} = \frac{1}{M} \sum_{c=1}^M \left(\sum_{i=1}^{N-1} |x_{c,i+1} - x_{c,i}| \right) \quad (2)$$

3. Zero crossings (ZC) A proxy for muscle motor unit firing frequency, measuring how often the signal crosses the zero-voltage baseline. $\mathbb{I}$ is an indicator function that returns 1 if true.

$$\text{ZC} = \frac{1}{M} \sum_{c=1}^M \left(\sum_{i=1}^{N-1} \mathbb{I}(x_{c,i} \cdot x_{c,i+1} < 0) \right) \quad (3)$$

3.2 IMU features

Calculated individually for all 24 IMU axes (accelerometer and gyroscope).

1. Mean and standard deviation (Std)

$$\text{Mean} = \mu = \frac{1}{N} \sum_{i=1}^N x_i \quad \text{Std} = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2} \quad (4)$$

2. Jerk Calculated as the standard deviation of the first derivative of the signal (the difference between consecutive samples). Useful for quantifying the explosiveness or smoothness of movement. Let $\Delta x_i = x_{i+1} - x_i$, and μ_Δ be the mean of those differences.

$$\text{Jerk} = \sqrt{\frac{1}{N-1} \sum_{i=1}^{N-1} (\Delta x_i - \mu_\Delta)^2} \quad (5)$$

3.3 Goniometer targets

The ground truth joint angle for a given window is simply the mean of the goniometer signal over that window.

$$\text{Target} = \frac{1}{N} \sum_{i=1}^N x_i \quad (6)$$